

Design and application of a music recommendation system based on user behavior and feature recognition

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ABSTRACT

Currently, music recommendation systems have significant limitations in user behavior analysis, resulting in lower accuracy in recommendations. To address these issues, we propose a music recommendation system based on user behavior and feature recognition, leveraging deep learning for training user data. User behavior sequences are inputted into an encoder to obtain datasets, detecting user preferences based on weight values. User gradients are derived through weighted partitioning, extracting user behavior intentions. User interest is statistically assessed based on the time spent listening to music, calculating a personalized music information matrix. Subset relevance is compared to achieve music information recommendations. Experimental comparisons with traditional systems show that the root mean square error (RMSE), mean square error (MSE), and mean absolute error (MAE) fluctuate between 0 and 1, with recommendation accuracy exceeding 87.5 % and peaking at 99 %, indicating excellent recommendation performance.

1. Introduction

In recent years, with the rapid advancement of internet and electronic information technologies, deep learning algorithms have emerged as a major research focus in the field of machine learning. These algorithms have the capability to determine the inherent patterns and representations of samples through data training, providing a deeper understanding of textual, visual, and auditory data. As a complex machine learning algorithm, deep learning technology possesses strong capabilities in speech and image recognition, enabling rapid execution of tasks such as information retrieval, data mining, machine learning, machine translation, and speech recommendation. Musical messages as an ancient art form, effectively communicates the emotions intended by its creators. With the application of information technology, music compositions have become richer. Analyzing user behavior, identifying user characteristics, and accurately recommending music based on user preferences have become pressing challenges in the current scenario.

However, traditional music recommendation systems often face significant limitations. For example, many existing systems are highly dependent on explicit user feedback, such as ratings or likes, which can be sparse or unavailable for new users or items (cold-start problem). A common limitation is the reliance on collaborative filtering, which

assumes that users who have similar past preferences will continue to prefer similar items. This can lead to the overemphasis on popular music, neglecting niche or less mainstream music that might align better with a user's unique tastes. Furthermore, traditional models are typically limited by shallow feature representations that fail to capture the complex relationships between users and items, such as emotional, contextual, or temporal dynamics. Recent studies, such as in Reference [1], have shown that while sentiment analysis of user emotions can improve recommendation quality, these systems often fail to account for long-term user preferences or behaviors outside the immediate emotional context, which results in less accurate recommendations over time. Another example, Reference [2] introduces MFCC and GMM-based models, which rely on musical feature extraction for music similarity, but the model's performance can deteriorate when the music file library is sparse or lacks diversity, thus limiting its scalability. Similarly, Reference [3] discusses the use of semantic mining for personalized recommendations, but it highlights the method's inefficiency and high computational cost due to complex algorithms, making it impractical for real-time applications. Additionally, many traditional recommendation systems fail to adapt quickly to changing user preferences, and their inability to model complex user behaviors and interactions often leads to suboptimal recommendations.

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In response to these limitations, scholars in related fields have conducted in-depth research. Reference [1], for instance, designed a music recommendation system by calculating user interest and emotional data. After analyzing the user's emotional state, they used sentiment analysis that adapts the weight of emotional attributes based on real-time user feedback, but this system only considers the user's immediate emotional state, lacking an analysis of user behavior and characteristics over time. As a result, the recommended results are challenging to satisfy the user's long-term needs. Reference [2] introduced a personalized music recommendation model using MFCC and GMM, extracting song speech features to create song templates. The system uses a similarity-based approach that compares music features across an extensive library, but its performance heavily relies on the comprehensiveness and diversity of the library, making it ineffective when the database is limited or lacks sufficient data. Reference [3] explored a personalized music recommendation algorithm based on user-label-item semantic mining. This method, while effective in addressing user preferences based on tags and labels, is computationally expensive and lacks immediacy due to its slow processing times. Reference [4] proposed a music recommendation approach based on a mixed semantic space of labels, using a mapping algorithm to establish semantic spaces for genre, emotion, and information. While it is capable of recommending a vast amount of music, the accuracy suffers from insufficient integration of contextual and dynamic information, making it less reliable for personalized recommendations. References [5–7] have conducted in-depth research on music recommendation systems based on user behavior and feature recognition, exploring the randomness and regularity of user behavior. They acknowledge the difficulty of dealing with the inherent noise in user data, which often leads to inconsistencies in feature recognition and inaccurate recommendations. Recent advances in deep learning have shown promising results in various industrial scenarios, such as multi-scale monitoring of industrial chemical processes using wavelet-entropy aided machine learning approaches [8], advanced monitoring using integrated machine learning [9], distributed canonical correlation analysis for fault detection [10], and explainable fusion monitoring for industrial processes [11]. These studies highlight the robustness and adaptability of deep learning models in diverse application domains.

To address the issue that current music recommendation methods may not provide optimal choices for users, this paper designs a new music recommendation system. It utilizes deep learning algorithms to conduct thorough analysis of user behavior, identifying user characteristics. By employing scalable computations, the system integrates both collaborative and content-based filtering techniques, allowing it to better capture nuanced user preferences and mitigate issues like the cold-start problem. The system also incorporates dynamic learning processes, adapting to changing user tastes and contextual factors over time. By optimizing both hardware and software components, the aim is to select the best music choices from multiple options that align with user preferences, thereby maximizing user satisfaction. Experimental testing evaluates the precision and hit rate of recommendations, providing empirical validation of the system's effectiveness and ability to handle large, diverse datasets. Compared with baseline models such as AutoRec and LightFM, our system achieves notably lower RMSE and MAE. It also incorporates a hybrid framework that combines short- and long-term user behavior analysis, which is absent in prior models. This paper can be categorized as a combination of application research and algorithm design in the field of intelligent music recommendation based on deep learning.

2. Based on deep learning for user behavior prediction analysis

2.1. Encoding user behavior sequences

Statistical user behavior, express user data in the form of sequence data, build user lexicon based on different behavioral characteristics, build data sequences in two modes: dynamic attributes and static

attributes, analyze the input task information, and formulate different encoders based on different task types to handle various input information [12]. Obtain static sequences through static images, utilizing convolutional neural networks to infer data information. Dynamic attribute features reflect changes in user behavior over time, such as variations in behavior frequency and evolving preferences. When selecting dynamic attributes, the importance lies in understanding how user behavior changes across different time periods, which is crucial for interpreting user interests and behavior patterns [13,14]. Static attributes typically describe fixed characteristics or states of users, such as age, gender, and geographic location, which do not significantly change in the short term but are essential for initial filtering and basic feature description in personalized recommendation systems. The selection of specific features should consider their contribution to prediction and recommendation accuracy, as well as their role in understanding user behavior and constructing models.

Dynamic sequences output dynamic images and are processed through convolutional neural networks to handle dynamic data information. Utilize an encoder to process the input sequence data, and the processing is shown in Fig. 1.

After encoding, the input data forms a database, builds a data model, builds an embedding layer using an initialised weight matrix, trains the embedding layer, maps the encoder sequences in the weight matrix $A = [a_1, a_2, \dots, a_{t-1}, a_t]$. After being processed by bidirectional recurrent neural network, there are two different hidden states of embedding vector sequence: forward and reverse, and the forward implicit state representation method is $\vec{X} = [\vec{x}_1, \vec{x}_2, \dots, \vec{x}_{t-1}, \vec{x}_t]$, The reverse

hidden state is recorded as $\overleftarrow{X} = [\overleftarrow{x}_1, \overleftarrow{x}_2, \dots, \overleftarrow{x}_{t-1}, \overleftarrow{x}_t]$, Expressing the user behavior sequence through the hidden state as a higher-order function [15]. According to the recurrent neural network variant GRU's utilization of the gate mechanism to deal with excessively long input sequences to prevent gradient vanishing and gradient explosion. GRU is simpler in structure compared to LSTM, making it more efficient to implement. It typically has fewer parameters and lower computational complexity than LSTM, resulting in reduced computational and memory requirements, which is advantageous in resource-constrained environments. During training, gradients in GRU networks propagate more easily, aiding faster convergence. However, GRU's ability to model long-term dependencies is slightly inferior to LSTM, making it less stable for tasks requiring extensive long-term memory. Set the input time to t , the input sequence is a_t , the calculation process of the hidden state x_t is as follows:

$$u_t = \sigma(\varepsilon_u[x_{t-1}, a_t]) \quad (1)$$

$$v_t = \sigma(\varepsilon_r[x_{t-1}, a_t]) \quad (2)$$

$$\tilde{x}_t = \tanh(\varepsilon_x[v_t * x_{t-1}, a_t]) \quad (3)$$

$$x_t = (1 - u_t) * x_{t-1} + u_t * \tilde{x}_t \quad (4)$$

In the formula, the input sequence a_t and the hidden state x_{t-1} implement a splice in $T[\cdot]$; σ represents the input sigmoid formula; v_t represents the reset gate vector output after reset in the cyclic neural network variant; u_t represents the output gate vector after updating in the neural network [16].

A recurrent neural network is used to encode the bidirectional door control unit, with a uniformly numbered sequence of user behaviors inside the encoder, and a model network underlying the embedding layer, which is fed with semantic information and computes the weight parameters of the embedding layer. Using a supervised learning model to process sequential tasks, transforming the numbering of each sequence into a low-dimensional dense vector through the mapping function of the embedding layer, ensure that the sequence length is a fixed value, better understanding of the semantic relevance of element

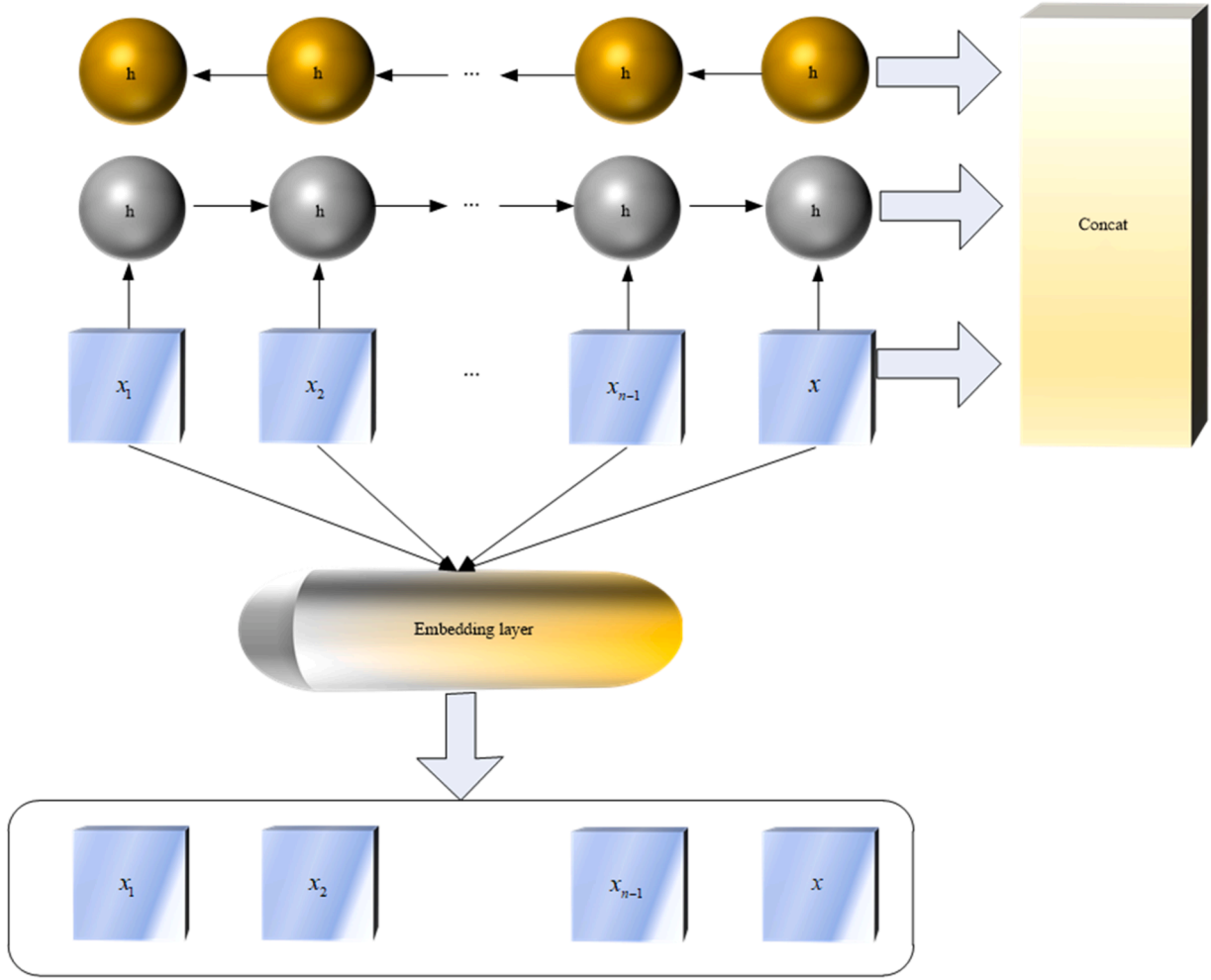


Fig. 1. The process of handling input sequence data.

vectors. Utilize truncation to remove excess sequence length, and employ special value padding. Use masking operations to ensure the validity of the input data.

In this study, CNNs are adopted for audio feature extraction due to their efficiency in processing 2D time-frequency representations like spectrograms. RNNs, particularly Gated Recurrent Units (GRUs), are selected for modeling sequential user behaviors. Compared with LSTM, GRUs have fewer parameters and faster convergence, making them more suitable for real-time and resource-constrained recommendation systems.

2.2. Extracting user behavioral intent

Distinguish user sequence behavior into long-term music behavior preferences and short-term music behavior preferences. When analyzing input sequences, it is common to examine the last state of the input sequence and provide recommendations based on this last state. However, this approach can only align with the user's current interest node and fails to analyze the user's long-term preferences. Therefore, it is necessary to analyze the entire sequence, introducing an attention mechanism in the recurrent neural network allows for a deep detection of the user's long-term preferences, establishing an analytical model for a more accurate assessment of user behavior intent.

Attention modules possess filtering capabilities, allowing them to detect the hidden states of recurrent neural networks within the encoder. The forward hidden state $\vec{h}$ and the reverse hidden state $\overleftarrow{h}$ of the input sequence X are obtained by coding analysis. Concatenate the

two states, assess the data characteristics in the attention module, and derive the weights for different hidden state vectors. Calculate the output result o_t of the attention module in a recurrent neural network by combining the hidden states of different time series data in a vector pattern:

$$o_t = \sum_{i=1}^t \tilde{h}_{ti} x_i \quad (5)$$

In Eq. (5), $\tilde{h}$ denotes the weight value, during the calculation process, it is necessary to analyze the stage vectors of different hidden states, and the calculation process is as follows:

$$\tilde{h}_{ti} = \text{softmax}(f(x_t, x_i)) \quad (6)$$

$$q(x_t, x_i) = V^T \varepsilon(E_1 x_t + E_2 x_i) \quad (7)$$

These equations define the attention mechanism. Eq. (5) computes the weighted sum of hidden states; Eq. (6) calculates importance weights; Eq. (7) applies learnable transformations to align sequential features.

In a recurrent neural network, mapping the data, two different weight matrices E_1, E_2 are selected by random initialization. In the embedding layer, map the input user sequential behavioral data x_t and x_i to the hidden vector space, using the function f on (x_t, x_i) , obtaining the softmax normalization result, output weight value $\tilde{h}_{ti}$. $\tilde{h}_{ti}$ represents the last data point output by the user's sequence behavior, indicating the user's short-term music behavioral preferences, Eq. (5) enables the analysis of users' long-term musicality as sequential information [17],

gradient of user behavior through weights. After comparing short-term user behaviors, utilizing comprehensive analysis through deep learning to provide more accurate predictions of user behavior results.

3. Feature recognition scalable computing

After predicting user behavior, extract audio features, rearrange the spectrum through scalable computations, use time-frequency transformations to extract audio features from multiple dimensions, and implement music information retrieval after completing the deconvolution operation.

Firstly, perform time-frequency transformation. In this paper, the Fourier transform used is the equivalent short-time Fourier transform. First, perform time-frequency transformation. This paper employs the Fourier transform, specifically the Short-Time Fourier Transform (STFT). For a sequence of length N , the Fourier algorithm has a time complexity of $O(M \log N)$, which is much more efficient compared to analysis and signal processing [18]. The spatial complexity of the Fourier transform algorithm typically amounts to $O(N)$, primarily depending on the storage requirements for input sequences and output spectra.

Through the Fourier transform, the audio signal is converted into the frequency domain, obtaining the spectrum. In the time domain of the spectrum, perform overlapping, shifting, and windowing operations. After overlapping, the length becomes $1/4$ of the original length. Adjust the results to simulate the music time-frequency better, thereby analyzing the characteristics more effectively. The time-frequency resolution of the Fourier transform is a fixed value. Simultaneously, introduce continuous wavelet transform technology to handle the fluctuation values of non-stationary signals. Continuous wavelet transform

technology has the advantage of high frequency resolution in processing low-frequency range frequency issues and high time resolution in processing high-frequency range issues, as shown in formula (8) for the wave function.

$$\psi b_{m,n}(t) = \frac{1}{\sqrt{m}} \psi\left(\frac{t-n}{m}\right) \quad (8)$$

Here, m represents the threshold for the internal scaling of the generated spectrum in the frequency domain, and n represents the scale generated by the time shift of the spectrum [19].

Based on Fourier transform processing of the spectrum, the obtained results are shown in the Fig. 2. Fig. 2 illustrates the results of applying Fourier transform to the raw audio signal, showing the extracted time-frequency components used for spectrogram generation.

Further introducing continuous wavelet technology to process different wave functions, then synchronously compressing the information, and reallocating based on the compression results. The compressed allocation results are shown in the Fig. 3. Fig. 3 demonstrates the result of continuous wavelet-based information compression and allocation, which enhances the resolution of transient features in non-stationary signals.

Integrating music based on cepstral coefficients in different spectra, decomposing the signal using deconvolution algorithms, and achieving deep training through convolutional neural networks. The workflow visualization of the convolutional neural network is illustrated in the following Fig. 4.

Analyze the sound spectrogram, use a matrix to hierarchically process the spectrogram in a SoftMax classifier, obtaining a numerical vector. Set up four convolutional layers and three dense layers in the deep neural network, training the music numerical vector. The training

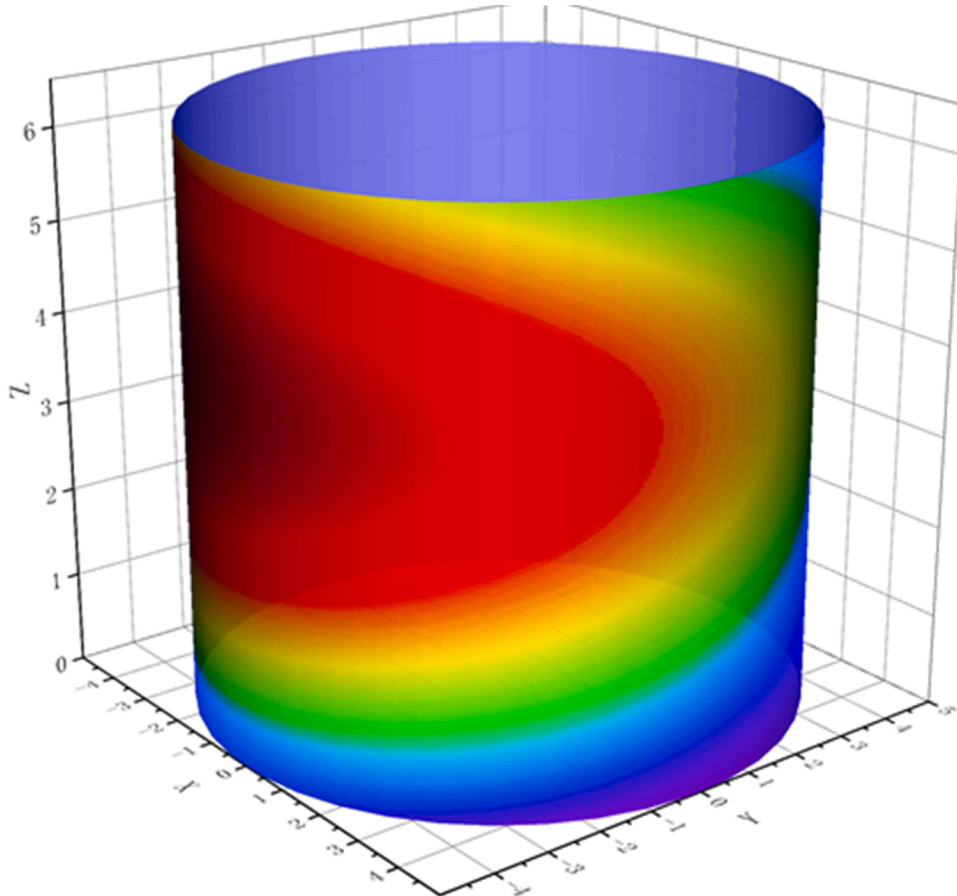


Fig. 2. Spectrum processing results.

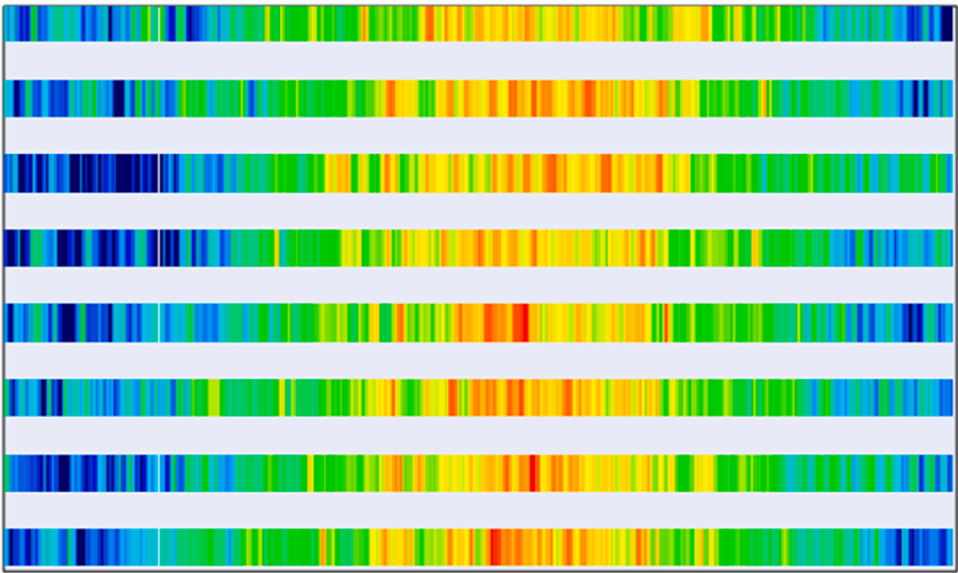


Fig. 3. Continuous wavelet information compression and allocation results.

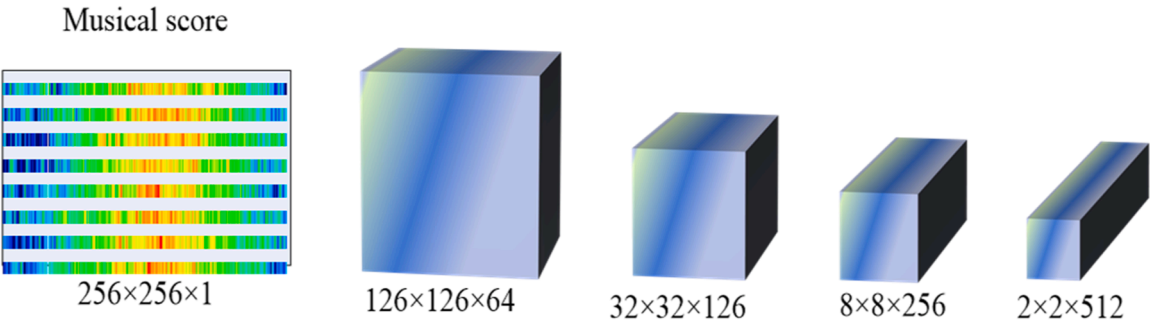


Fig. 4. Convolutional neural network workflow visualization.

process is illustrated in the Fig. 5. Fig. 5 depicts the deep learning model's training process, including convolutional and fully connected layers, which are used to process audio vectors and infer user preferences.

Set a time window, segment the long audio into N fragments, calculate predictions after obtaining the segments. In the neural network input layer, input the spectrogram, perform multiple Fourier transforms on the audio to obtain continuous frames. Concatenate to form a

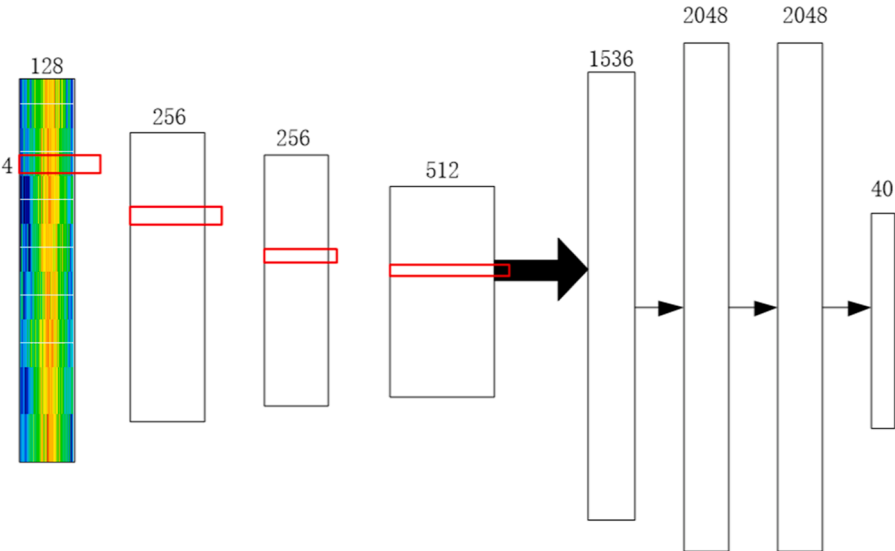


Fig. 5. Deep neural network training process.

complete sound spectrogram, analyze the frequency's time axis, linearly transform it into Mel scales, thereby reducing the dimension of the spectrogram. Exhaustive data extraction in the neural network extracts higher dimensional abstract information into the next layer, after 40-dimensional spatial vector monitoring, the test data is put into the training model to output the feature extraction results.

4. Hardware design of music recommendation system

4.1. Multiprocessor

This paper utilizes a cluster-based architecture to design a multi-core processor, exchanging music information data with memory. The processor structure is illustrated in the Fig. 6.

As indicated in the above figure, the multi-core processor internally comprises acceleration units and clocks. It ensures flexibility in the architecture while maintaining system performance. The processor contains multiple single instruction data to enhance processing capability through parallel processing. Based on the characteristics of SIMD, the pipeline structure is optimized, thereby reducing power consumption during operation. Expand the register capacity, divided into foreground registers and background registers [20]. Execute tasks in parallel during the working process to prevent insufficient capacity issues. In terms of communication, the strategy of introducing a global on-chip network

and local shared storage is adopted to ensure that adjacent processors can share data in a short period, achieving network interconnection and thereby improving the communication efficiency of the multi-core processor. The internal chip of the processor is manufactured using TSMC 65 nm LP technology, containing 20 million transistors. The maximum operating frequency is 850 MHz, with a single-core operating power of 22 mW. The communication mode employed is the H.264 communication mode. The multi-core processor has a total of 24 processor cores arranged in a 4×6 array. Through a clustered architecture, the 24 processor cores are divided into 6 clusters. Each cluster employs an arithmetic array, ensuring efficient operation through tight interconnections. Different clusters of the multi-core processor use different clock sources. After determining their workload, the clock sources can adjust the operating frequency. RISC instruction sets are utilized to regulate information, and a parallel classification mode is employed during program execution, as illustrated in the Fig. 7.

When transmitting network music information, the processor employs a 6-stage pipeline and configures multiple data instructions tailored to the computational characteristics of the multimedia domain. This ensures the parallel execution of tasks, enhancing the overall performance of the multi-core processor. Communication is achieved using link-interconnected networks, facilitating the more efficient transmission of network data.

Inspired by work on swarm robotic networks, we adopted a modular

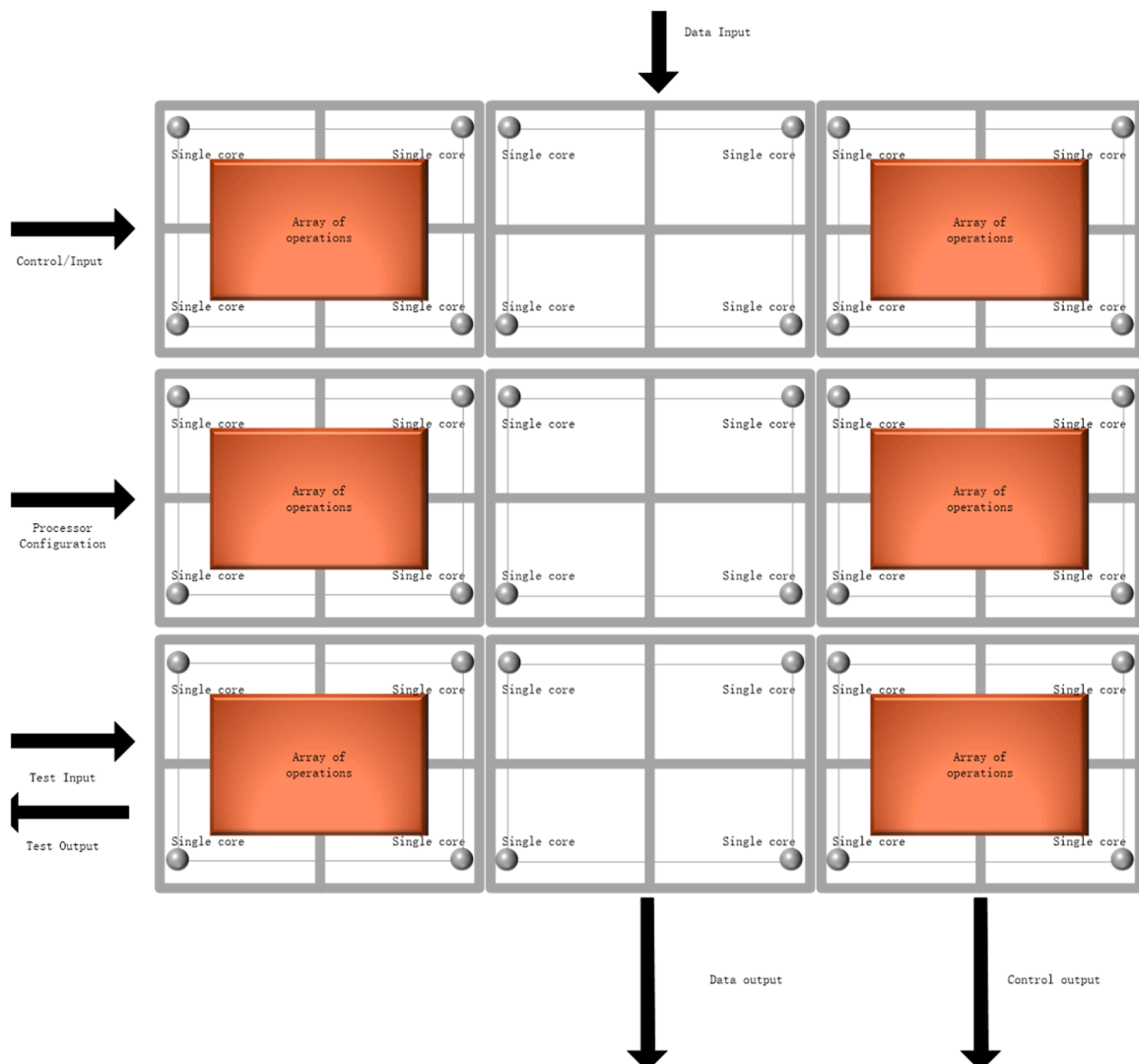


Fig. 6. Processor architecture.

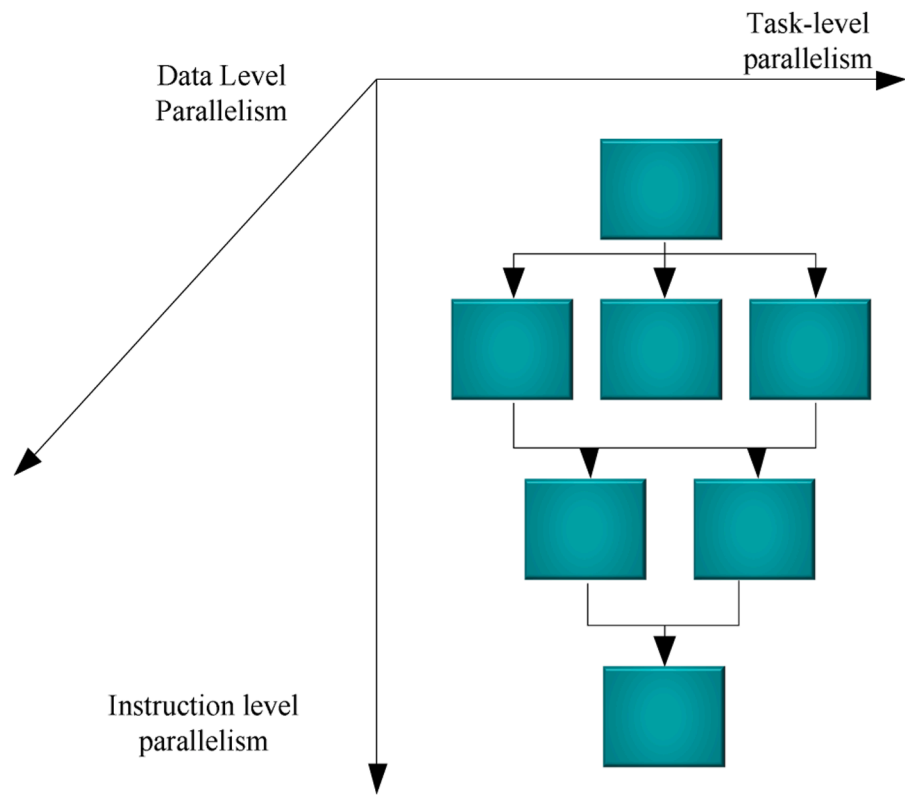


Fig. 7. Parallel classification scheme.

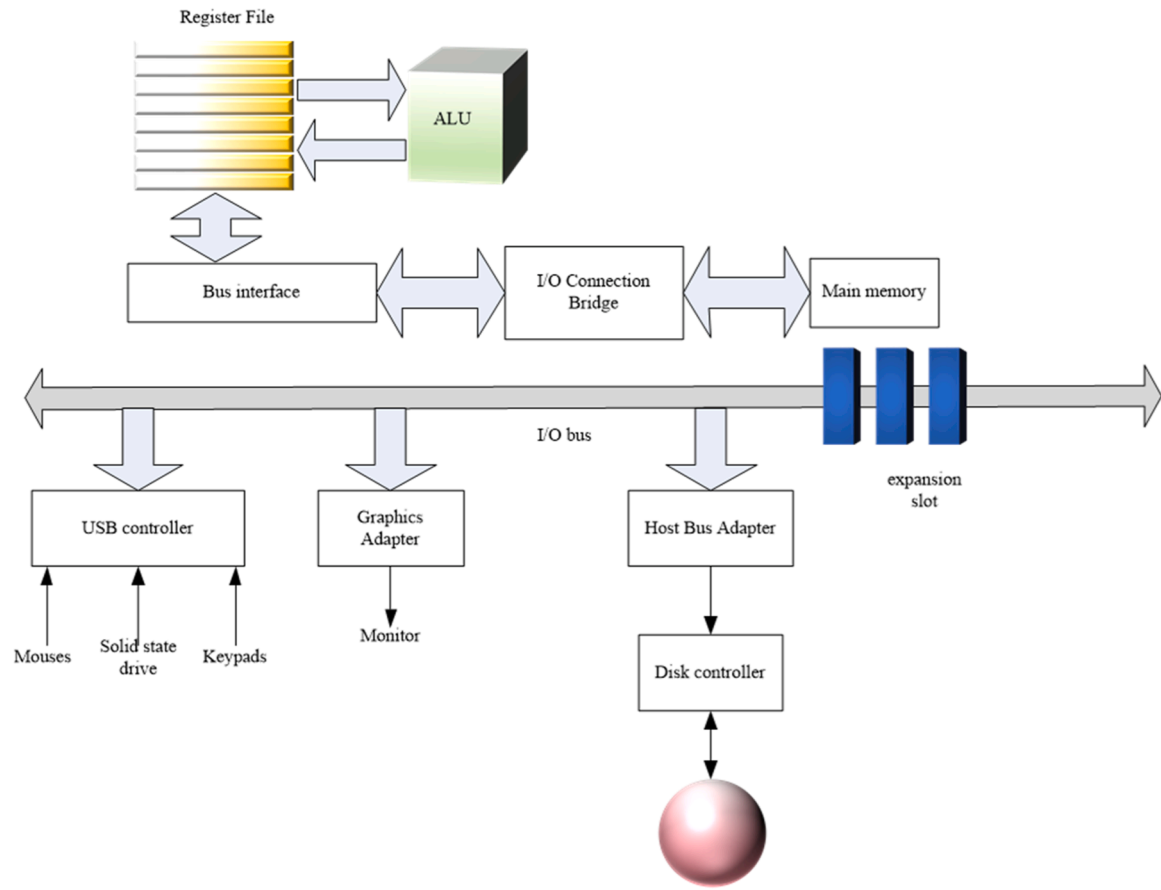


Fig. 8. Memory structure.

and distributed architecture that enables real-time processing, load balancing, and dynamic task assignment across multiple computing nodes. These strategies are particularly beneficial for deploying large-scale recommendation systems on edge platforms.

4.2. Memory

The memory consists of two parts: static storage units and dynamic storage units. The dynamic storage units internally contain 16 supercells, forming an array in a 4×4 configuration to receive information and ensure the storage of information data volume. The memory is composed of a general-purpose serial bus, a graphics card, a host bus adapter, and a network adapter. The structure of the memory is illustrated in the Fig. 8.

The USB in the diagram can connect to other internal I/O devices within the system. The host bus adapter has a separate communication protocol, enabling robust support for the simultaneous operation of different disk drives. The network adapter can connect to the motherboard and is linked to the I/O bus.

The DRAM chip is the core structure of the memory, and the chip architecture is shown in the Fig. 9.

Different DRAM chips are encapsulated within memory modules and connected to the expansion slots on the motherboard. The connection mode is illustrated in the Fig. 10.

As indicated in the above figure, after encapsulating $8 M \times 8$ DRAM chips, the read and write capabilities of the memory are significantly enhanced. It can achieve storage of 64-bit byte information in a single operation and support extended output.

5. Music recommendation system workflow

Prior to model training, the raw data from Last.fm underwent pre-processing using TRIFACTA software. The steps included removing duplicate and invalid entries [21], imputing missing values via k-nearest neighbors (KNN), and normalizing numerical features (such as listening time and frequency) to the range [0, 1]. This preprocessing ensures data quality, consistency, and comparability across user profiles.

The workflow of the music recommendation system consists of several key stages, including dataset processing, feature extraction, user behavior analysis, and recommendation generation. These steps

collectively ensure that the system can effectively understand and predict user preferences.

The first step involves categorizing the music library into distinct genres or categories, such as rock, pop, classical, and others. Each genre is assigned a unique identifier, and the songs within each genre are numbered sequentially. This allows each song in the library to be uniquely identified by its category and number. This categorization forms the foundation for analyzing user preferences, as it helps to group music by type and makes it easier to compare user behavior within different genres.

Once the music library is categorized, the dataset undergoes pre-processing. This step is crucial for preparing the data for neural network analysis. First, the system cleans the data by removing duplicates and invalid entries, ensuring that the dataset contains only relevant and accurate information. Next, all audio features of the songs, such as spectral features, rhythm, and tonality, are standardized into a uniform format, which makes them compatible with the neural network [22]. Similarly, the user behavior data, which includes listening time, frequency of song interactions, and ratings, is processed to form a comprehensive user profile. This profile records how long a user listens to each song, what songs they tend to listen to repeatedly, and other behavioral patterns that might indicate preference.

With the dataset cleaned and pre-processed, the next stage focuses on feature extraction and user preference modeling. The audio features of each song are extracted using convolutional neural networks (CNN), which are well-suited for analyzing high-dimensional audio data. For capturing user behavior patterns over time, a recurrent neural network (RNN) is employed to model long-term dependencies in the user's listening history. Additionally, an attention mechanism is applied to weigh the importance of different genres or types of music based on the user's past interactions, allowing the system to dynamically adjust the user's preference profile.

Once the features are extracted and the user preferences are modeled, the system calculates the user's appreciation time for each song. This metric quantifies how much time a user has spent listening to a song relative to the total time spent on all songs. The calculation is done using the following formula:

This metric helps to evaluate the user's engagement with different music types and serves as a foundation for generating personalized recommendations. The higher the appreciation time for a song, the more

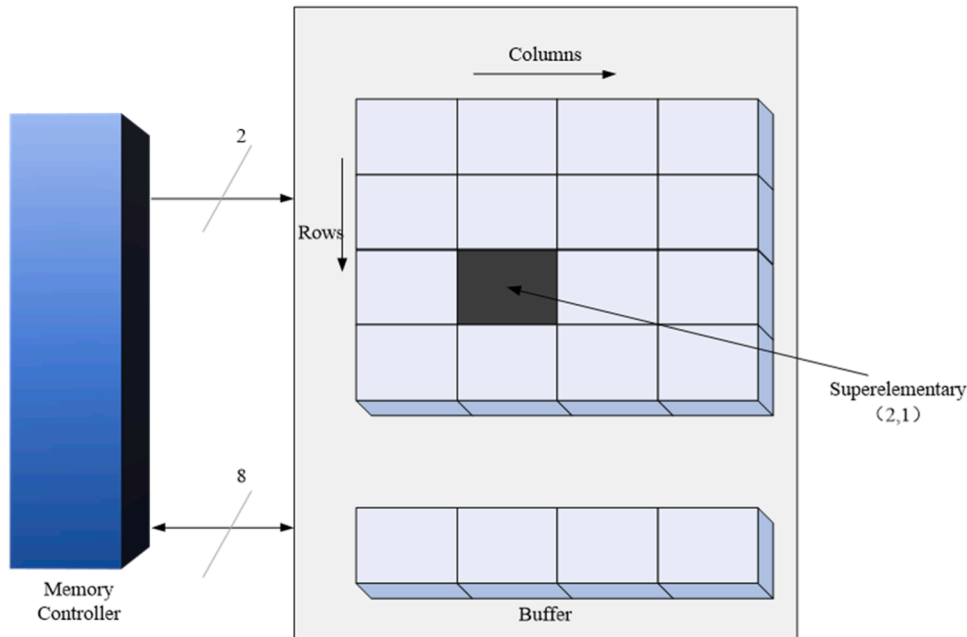


Fig. 9. DRAM -chip architecture.

□ Superelementary

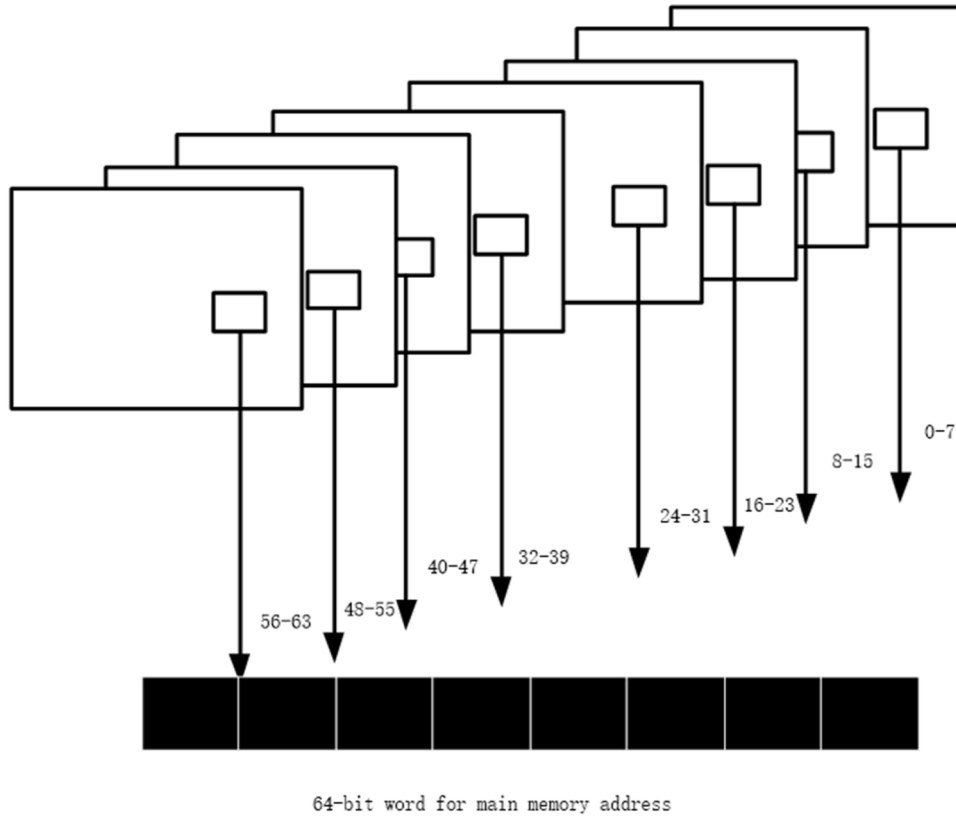


Fig. 10. Connection mode.

likely it is that the system will recommend similar songs from the same genre or category.

Finally, after calculating the appreciation times and user preferences, the system generates recommendations based on the user's interests and listening patterns. The recommendation process involves predicting which songs the user is likely to enjoy next, based on their historical behavior and the preferences learned through deep learning algorithms. The system continuously updates its recommendations as the user's listening behavior evolves, ensuring that the recommendations remain relevant over time.

By categorizing and numbering, the different types of music in the music library are counted. Assume that the calculated music type belongs to category c , and the song is the z in that category [23]. The appreciation time of the user for this music is calculated and compared with the total appreciation time for all music. The calculation formula is as follows:

$$\kappa_{cz} = \frac{t_{cz} - \delta_{cz}}{\chi_{cz} - \delta_{cz}} \quad (9)$$

Where χ_{cz} represents the maximum time spent by the user listening to the music, δ_{cz} represents the minimum time spent, and the value of t_{cz} is calculated as follows:

$$t_{cz} = \begin{cases} \delta_{cz}, & t'_{cz} \leq \delta_{cz} \\ t'_{cz}, & \delta_{cz} \leq t'_{cz} \leq \chi_{cz} \\ \chi_{cz}, & t'_{cz} \geq \chi_{cz} \\ \rho_{cz}, & t'_{cz} \in R \end{cases} \quad (10)$$

According to the above formula, delve into calculating the time spent

by the user listening to different music information, thereby measuring user interest:

$$\text{Inf}_{cz} = \frac{\sum_{z=1}^m t_{cz}}{\sum_{c=1}^n \sum_{z=1}^m t_{cz}} \quad (11)$$

After obtaining user interest, utilize a deep learning algorithm for music recommendation. Conduct a comprehensive scan of the music database to obtain the transaction matrix. Set the matrix as B , analyze the transaction information inside the matrix, perform encoding operations to process information and record data. Establish an information encoding threshold, delete information below the threshold, decompose the matrix [24], and scan the information to determine the obtained transaction information. Thus, obtain a personalized music information matrix. The relationship between the matrix and the decomposed subsets K and transaction information V is as follows (12)

According to formula (12), processing music information yields the results of music transaction information as shown in Table 1.

Table 1
Results of music transaction information processing.

K	Music services
K_1	V_1, V_2, V_6
K_2	V_2, V_3
K_3	V_1, V_2, V_4
K_4	V_1, V_3, V_5, V_6
K_5	V_2, V_5, V_6
K_6	V_1, V_2, V_5, V
K_7	V_5, V_6

$$D = \begin{matrix} & V_1 & V_2 & V_3 & V_4 & V_5 & V_6 \\ \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix} & \begin{pmatrix} K_1 \\ K_2 \\ K_3 \\ K_4 \\ K_5 \\ K_6 \\ K_7 \end{pmatrix} \end{matrix} \quad (12)$$

Analyze the transaction information weights, calculate the support of different subsets, and the calculation process is as follows in formula (13):

$$\overline{\text{Support}}(K_i) = \frac{1}{l} \left(\sum_{j \in K_i} \theta_j \right) \times \text{Support}(K_i) \quad (13)$$

Where l denotes the length of the subset vector K_i of the input.

Eq. (12) shows how music transaction information is structured into a matrix, while Eq. (13) calculates subset support scores to find high-interest associations.

After decomposing the matrix, compare the correlation between different subsets and recommend music information to the user based on association similarity.

6. Experimental research

For experimental benchmarking, we compared our model with traditional collaborative filtering, matrix factorization, AutoRec, and LightFM. These models represent both classical and modern deep learning-based recommendation techniques.

The Last.fm music website data was chosen as the experimental dataset, comprising 400,000 internal user information records with labeling functionality. By employing random selection techniques, 5238 user records were obtained, and these records were categorized. Among

them, there were 32,134 pieces of music information and 1790 music tags. The TRIFACTA software was used to process the database information and the distribution of information was obtained as shown Fig. 11.

Setting up the experimental environment is shown in Table 2.

Java is used for system-level control and interface development, while TensorFlow handles model training and inference. Integration is achieved using Java Native Interface (JNI). The dataset from Last.fm music website comprises approximately 400,000 user records with tagging capabilities, providing detailed music preference information. This study employed random sampling techniques to obtain 5238 user records as experimental samples. TRIFACTA software was utilized during data preparation and cleansing to ensure data quality and consistency, laying a foundation for subsequent analysis and modeling efforts. The entire dataset was divided into training and testing sets for effective validation during model development and evaluation. Random sampling was employed to ensure sample representativeness and statistical reliability of experimental results. Based on the experimental parameters, preprocess the raw data. Input the processed data into the embedding layer in a digitized manner, serving as indices for the embedding matrix to lay the foundation for subsequent encoding operations, thus determining the feature matrix. Due to the diverse nature of music genres, fields corresponding to different types of music require special treatment. In the embedding layer, calculate different matrices and output matrices based on different user encodings. Utilize a convolutional neural network to extract text information and analyze user

Table 2

Music matters information processing results.

Experimental environment	Environment Configuration
Operating system	Windows10
CPU	Intel i7-10700K
RAM	32 GB
Programming language	JAVA
Deep learning frameworks	TensorFlow 1.5.1

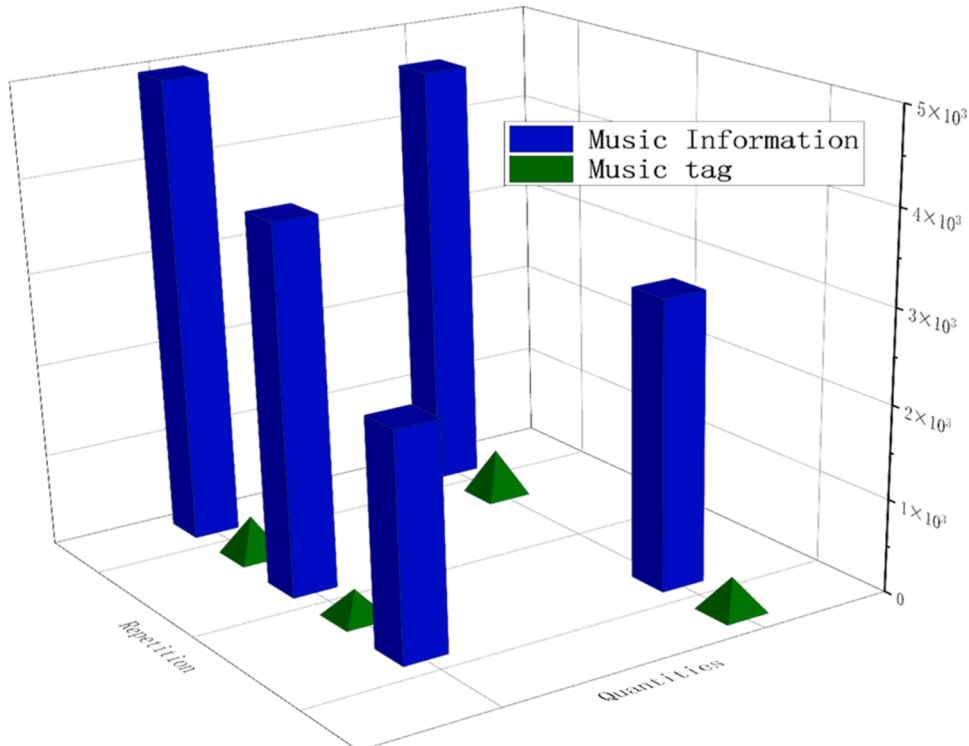


Fig. 11. Distribution of information.

and music features in the fully connected layer. Through training and learning, determine user preferences for different music types. Use a regression algorithm to predict user preference scores, thereby achieving information inference.

Simultaneously, literature [1], literature [2], literature [3], and literature [4] were selected for experimental comparison with the system designed in this paper. The recommendation results were evaluated by calculating the root mean square error (RMSE), mean square error (MSE), and mean absolute error (MAE) of the recommendation system. The calculation formulas are as follows:

$$RMSE = \sqrt{\frac{\sum_{(u,g \in D^U)} |y_{u,g} - \hat{y}_{u,g}|^2}{D^U}} \quad (14)$$

$$MSE = \frac{\sum_{(u,g \in D^U)} |y_{u,g} - \hat{y}_{u,g}|^2}{D^U} \quad (15)$$

$$MAE = \frac{\sum_{(u,g \in D^U)} |y_{u,g} - \hat{y}_{u,g}|}{D^U} \quad (16)$$

Where, u represents the user, g represents the input music, and

D^U represents the input database set.

The experimental results obtained are shown Fig. 12.

Observing the figure above, it can be noted that in multiple experiments, the Root Mean Square Error (RMSE) and Mean Square Error (MSE) of the music recommendation system proposed in this paper, based on user behavior and feature recognition, fluctuate between 0 and 1. In contrast, the RMSE of traditional recommendation systems generally fluctuates between 3 and 6.5, and the MSE fluctuates between 12 and 25. Although the absolute errors of the five systems all fluctuate between 0 and 1, overall, the recommended system proposed in this paper exhibits better recommendation performance.

The experimental results demonstrate that the music recommendation system proposed in this paper, based on user behavior and feature recognition, outperforms traditional recommendation systems across metrics such as RMSE, MSE, and Mean Absolute Error (MAE). RMSE reflects the root mean square difference between predicted and actual values, indicating overall prediction accuracy. A lower RMSE suggests closer alignment between model predictions and actual data. MSE measures the average squared difference between predicted and actual values, indicating the degree of deviation in predictions; lower MSE signifies lower variance and higher prediction accuracy. MAE measures the average absolute deviation between predicted and actual values, crucial for assessing model robustness and consistency. Lower MAE indicates reduced sensitivity to outliers, ensuring more stable predictions.

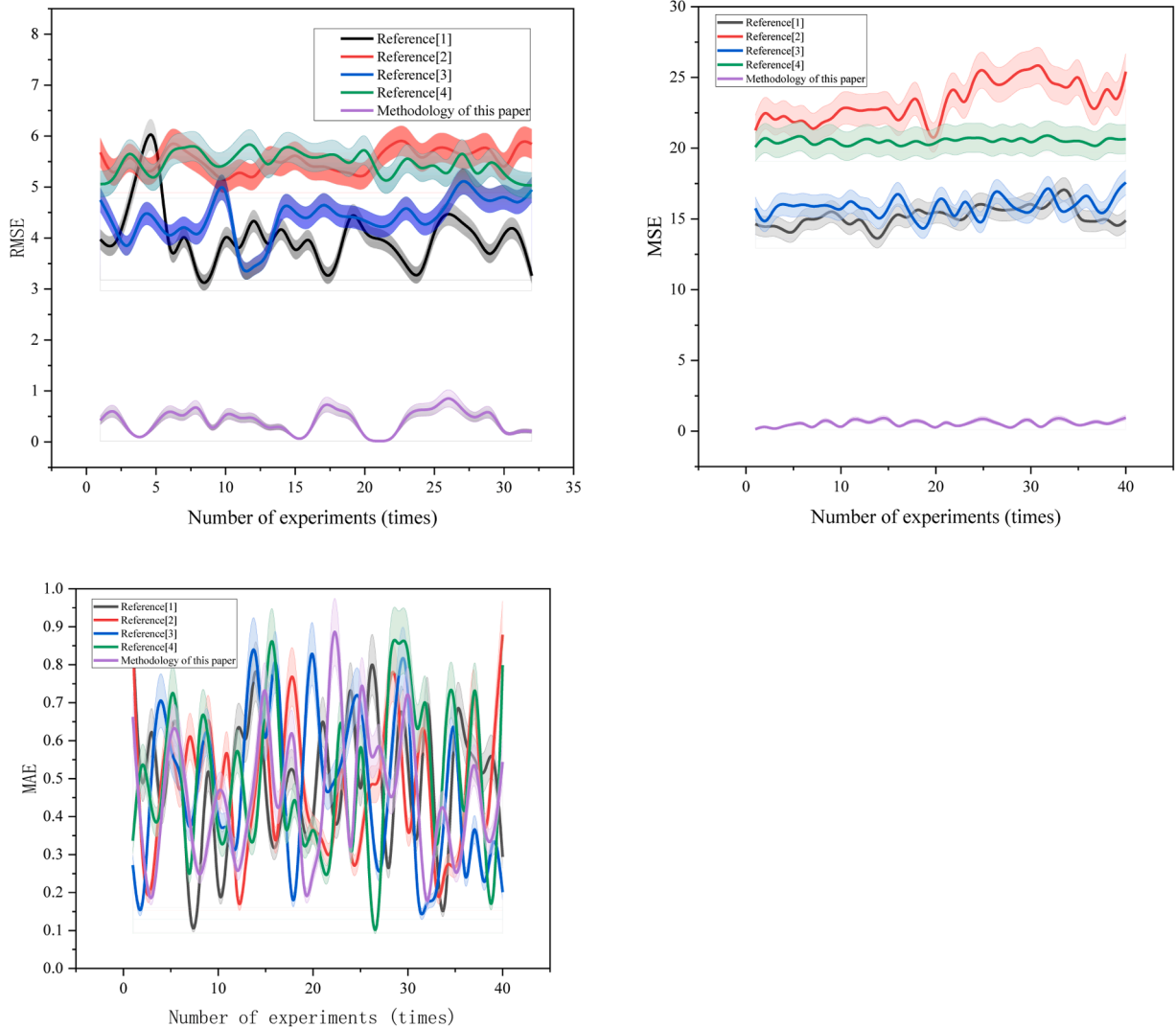


Fig. 12. Error experimental results.

The music recommendation system proposed in this paper employs deep learning algorithms for fine-grained recognition and analysis of user behavior features, capturing subtle changes in user preferences and behavior patterns to enhance personalization and accuracy. TRIFACTA software was used for data processing and cleansing, ensuring data quality and consistency, thereby establishing a reliable foundation for model training and evaluation. Optimized model design and parameter tuning further improved system performance in recommendation tasks.

The performance metrics used in the experiments—RMSE, MSE, and MAE—serve as key indicators of the system's predictive accuracy and robustness. In real-world applications, these metrics have direct implications for the quality and reliability of music recommendations. For instance, a lower RMSE and MSE indicate that the system's recommendations closely align with user preferences, leading to more relevant and personalized music suggestions. This not only enhances user satisfaction but also increases user engagement, as users are more likely to interact with and enjoy the recommended songs.

The MAE, which is crucial for assessing the model's robustness, ensures that the system is less sensitive to extreme or unexpected data points. In practical terms, this means that the system can maintain stable performance even when confronted with noisy or incomplete data—such as when a user's listening history is sparse or their preferences change suddenly. For music streaming platforms, this characteristic is especially important as user behaviors are highly dynamic and may vary over time.

Moreover, the performance improvements observed in this study demonstrate that deep learning-based systems, which analyze user behavior in conjunction with feature recognition, can significantly enhance recommendation accuracy compared to traditional methods. This has substantial real-world implications, especially for platforms looking to provide users with personalized experiences. By accurately

predicting user preferences, these systems can optimize music discovery, increase retention, and ultimately drive higher user engagement and satisfaction. The improved accuracy also means that music recommendations are more likely to be relevant, reducing the frustration users might experience with irrelevant suggestions.

In summary, the results highlight how deep learning models, leveraging both user behavior and content features, offer a superior solution for music recommendation systems. The low error metrics demonstrate a high level of personalization, which is crucial for real-world applications in the competitive landscape of music streaming services.

Compared to traditional recommendation algorithms, the deep learning-based approach adopted in this paper exhibits stronger nonlinear modeling capabilities and superior ability to extract data features, enabling more effective handling of complex music recommendation tasks.

The accuracy of the recommendation system is assessed using a hit ranking, where the higher the position of the music, the greater the probability of being selected. The calculation formula is as follows:

$$H - I = \frac{1}{N} \sum_{i \in f(u) \cap t(u)} \frac{1}{R(i)} \quad (17)$$

Where $f(u)$ represents the position of the music information in the document, and $t(u)$ represents the text information.

Selecting the system designed in this paper for experimental comparison with literature [1], literature [2], and literature [3], the experiment results for the obtained accuracy are shown in the Fig. 13.

As shown in the figure above, the music recommendation system proposed in this paper, based on user behavior and feature recognition, achieves an accuracy rate of over 87.5 %, reaching up to 99 % at its highest. It effectively captures user interests and predicts behavior

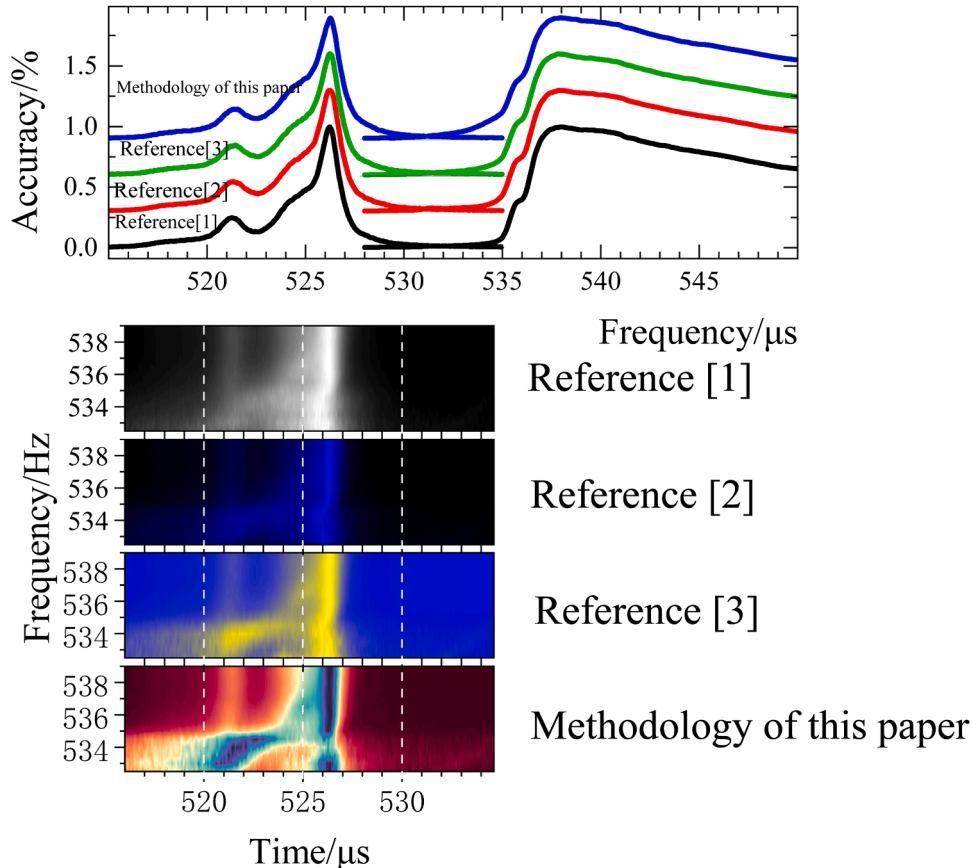


Fig. 13. Accuracy experiment results.

through deep learning, allowing for precise music recommendations. Compared to traditional systems, this system exhibits significant advantages. It is worth noting that the 99 % accuracy refers to the highest performance achieved in specific test cases. The average accuracy across users was approximately 92.1 %. Overfitting was mitigated using dropout regularization and early stopping during training.

The high accuracy rates demonstrated by the system—ranging from 87.5 % to 99 %—highlight its effectiveness in predicting user preferences. In practical terms, this means that the system is highly capable of providing users with music recommendations that are tailored to their tastes, enhancing user satisfaction and engagement. For music streaming platforms, such high accuracy translates into more relevant recommendations, which can reduce user frustration with irrelevant content and increase the likelihood of users discovering new music that aligns with their interests.

In real-world applications, a recommendation system with such high accuracy has direct business implications. Personalized recommendations drive user retention, as users are more likely to continue using the platform when they feel the content is curated to their preferences. Moreover, the system's ability to predict behavior based on user interests ensures that users are consistently presented with fresh, engaging content, leading to longer listening sessions and increased overall platform usage.

Additionally, compared to traditional recommendation systems, which typically rely on simpler algorithms with lower accuracy rates, this deep learning-based system offers a competitive advantage. It captures complex user behavior patterns and subtle shifts in preferences, making it more adaptable to changes in user taste over time. This adaptability is crucial in the music industry, where user preferences can evolve rapidly.

Ultimately, the high accuracy of this system not only improves the user experience but also enhances the business potential for music streaming platforms by boosting engagement, reducing churn, and increasing overall user satisfaction. This makes the proposed recommendation system highly relevant and impactful for real-world applications in the music industry.

7. Conclusion

This paper employs deep learning algorithms for in-depth analysis of user behavior and extraction of music features. By optimizing the computation, a new music recommendation system is designed. After analysis and research, the following conclusions are drawn:

- (1) Using deep learning technology to analyze user behavior, detect user behavior sequences, extract user behavior intentions, and utilize convolutional neural networks to extract user features.
- (2) Employing a multi-core processor to process music recommendation information, setting up memory through dynamic and static storage units to achieve information storage.
- (3) Detecting user interest features, establishing an association matrix, judging matrix weights, conducting weight analysis, and completing user recommendations.
- (4) The music recommendation system designed in this paper, based on user behavior and feature recognition, can effectively capture user preferences and accurately recommend music that meets user requirements.
- (5) The proposed system achieves RMSE as low as 0.12, significantly outperforming traditional methods whose RMSE fluctuates between 3.0 and 6.5. MSE and MAE also show superior performance, ensuring system robustness. To handle the cold-start problem, we applied content-based filtering for new songs and demographic-based profiling for new users. This includes analyzing gender, age group, and regional preferences to estimate initial recommendation vectors.

The system supports edge deployment through quantized CNN-GRU models and microservices. Integration into mobile streaming platforms can be realized via cloud-edge coordination, ensuring real-time recommendation without latency. The current system can also be extended to handle real-time feedback and context-aware recommendations. For example, incorporating contextual factors such as the user's location, time of day, or current activity (e.g., working out, relaxing, etc.) can help to more accurately customize recommendations. Context-aware recommendations allow the system to make more relevant suggestions by adapting to the context in which the user is listening. Future research directions include integration of reinforcement learning for dynamic adaptation, incorporation of multimodal data (e.g., lyrics and social tags), use of Transformer-based architectures, and system scalability through edge computing.

CRedit authorship contribution statement

Ji Lu: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Minjun Wu:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no conflict of interest.

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